Stepper Motor

Specifications:

- Speed and position can be controlled through data pins.
- Multiple coils.
- High torque (multiple gears inside)
- Low cost
- Long lasting
- Brushless operation



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Stepper Motor Basics

- Stepper motors use a cogged wheel and electromagnets to nudge the wheel round a ‘step’ at a time.
- Each high pulse sent energizes the coil, attracting the teeth closest to the cogged wheel and rotating the motor in precise and fixed angle increments known as steps.
- The number of steps that the stepper motor has in a 360 degree rotation is actually the number of teeth on the cog.

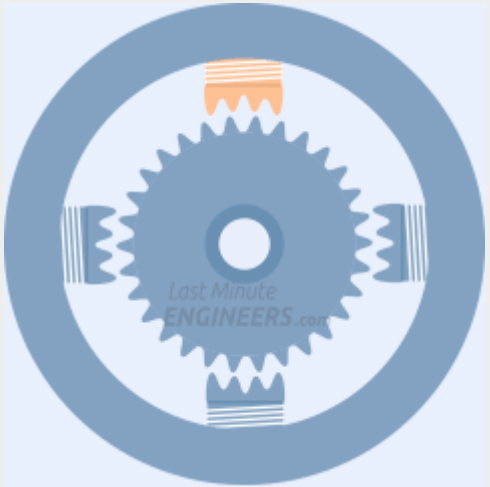
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Stepper Motor Basics

The way you pulse these coils determines how the motor operates:

- The sequence of pulses determines the spinning direction of the motor.
- The frequency of the pulses determines the speed of the motor.
- The number of pulses determines how far the motor will turn.

By energizing the coils in the correct sequence, the motor is rotated.



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Stepper Motor

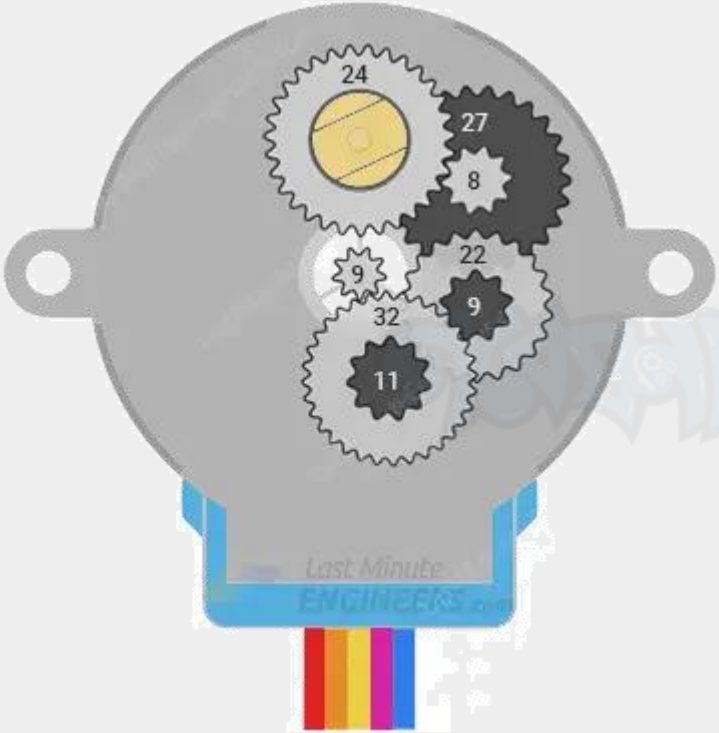
Specifications:

- Model: 28BYJ-48
- Power Rating 5v 240mA
- Phases: 4
- Gear Ratio: 64:1
- Step Angle: 5.625°/64



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Gear Ratio



Gear Ratios:

- 32 / 9
- 22 / 11
- 27 / 9
- 24 / 8

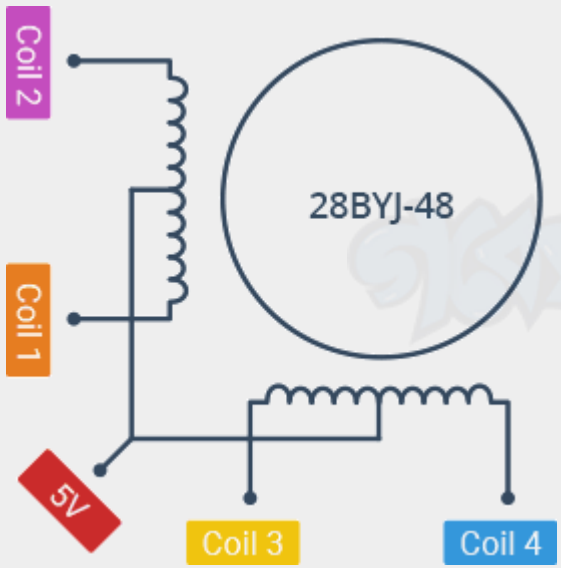
Multiplying the gear ratios:

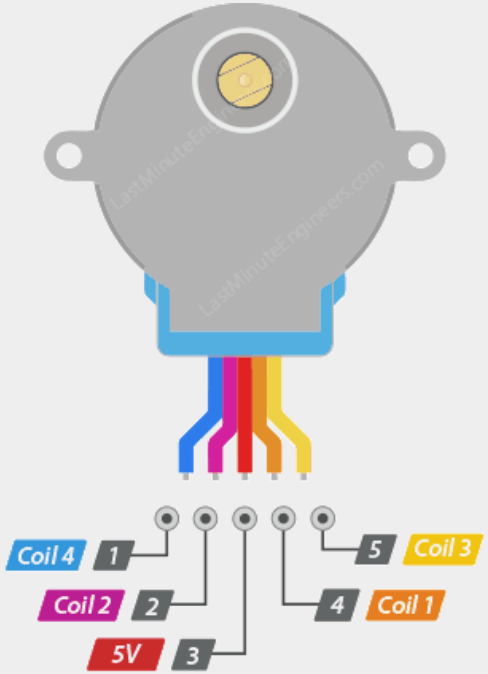
$$\frac{32}{9} \times \frac{22}{11} \times \frac{27}{9} \times \frac{24}{8} = 64$$

This gives us a **64:1** gear ratio

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Pinout



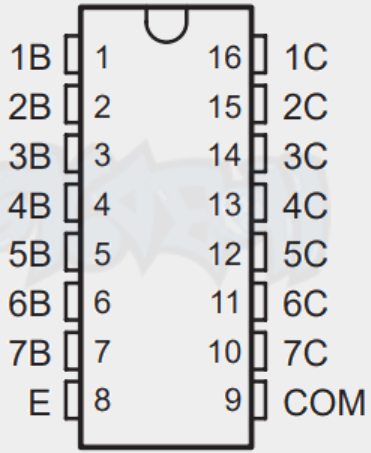


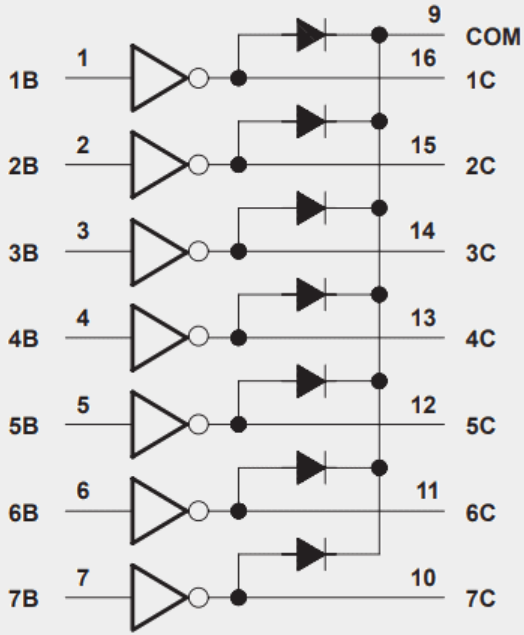
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Stepper Motor Driver

ULN2003

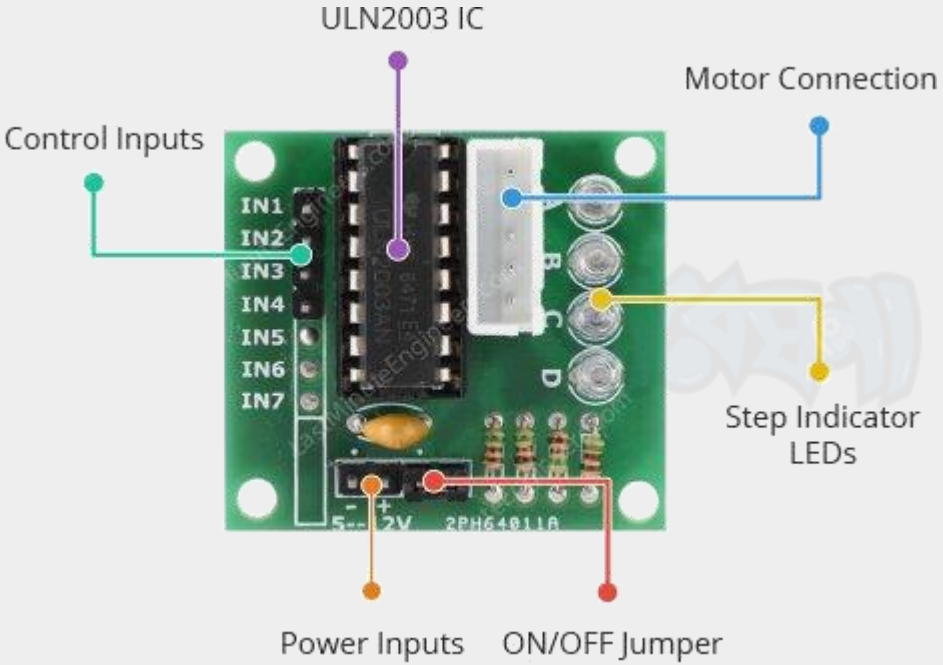
Max out: 500mA 50V

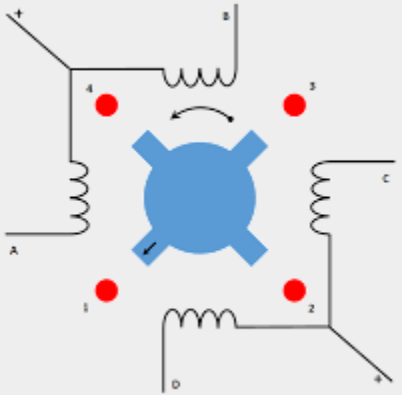




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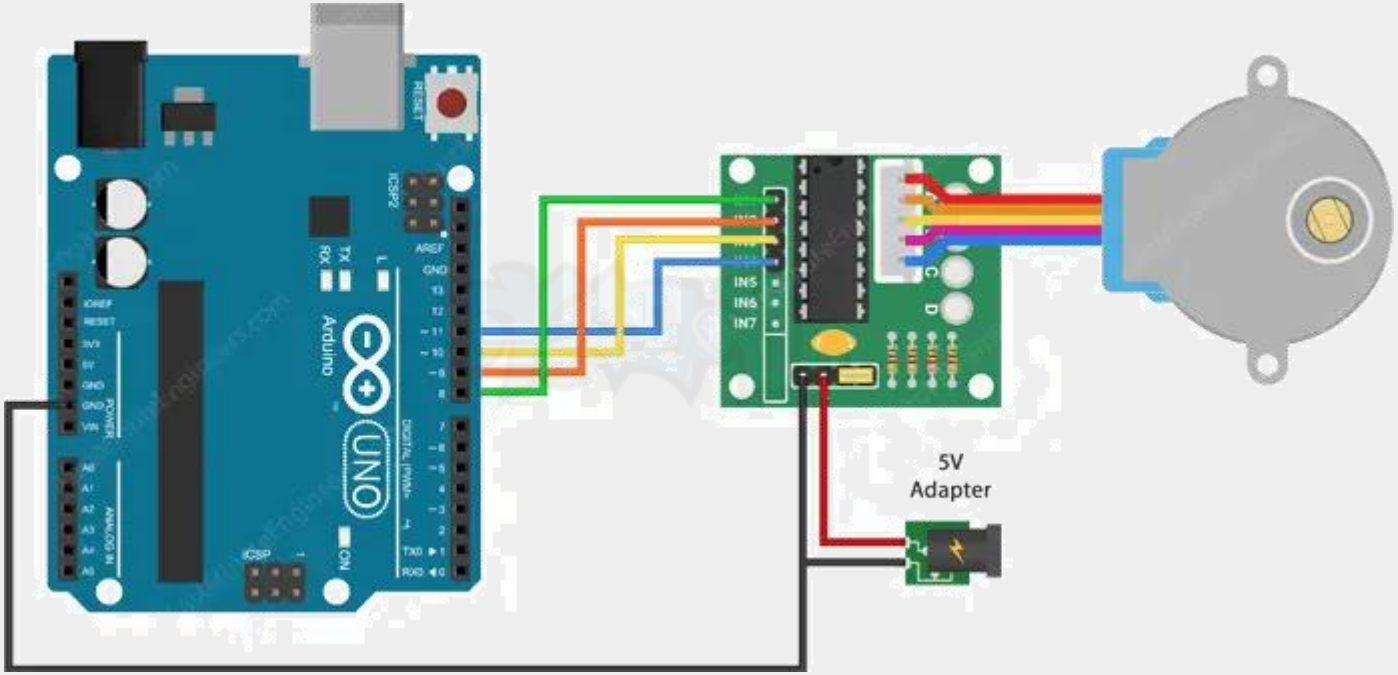
Pinout





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Sample Circuit



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Joystick

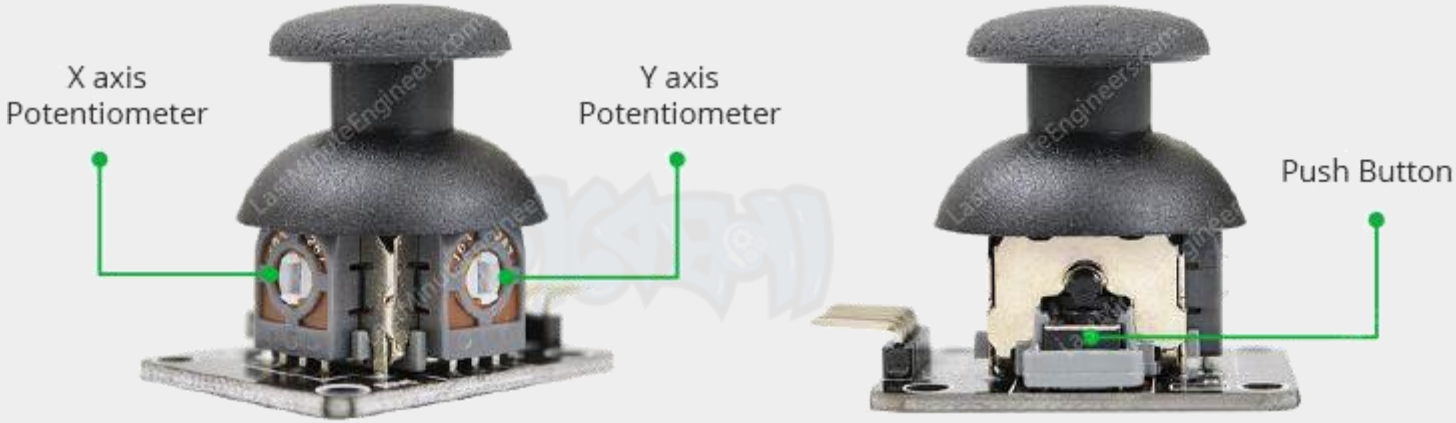
Specifications:

- 2 Axis
- 1 button
- Potentiometer based operation
- Low Cost



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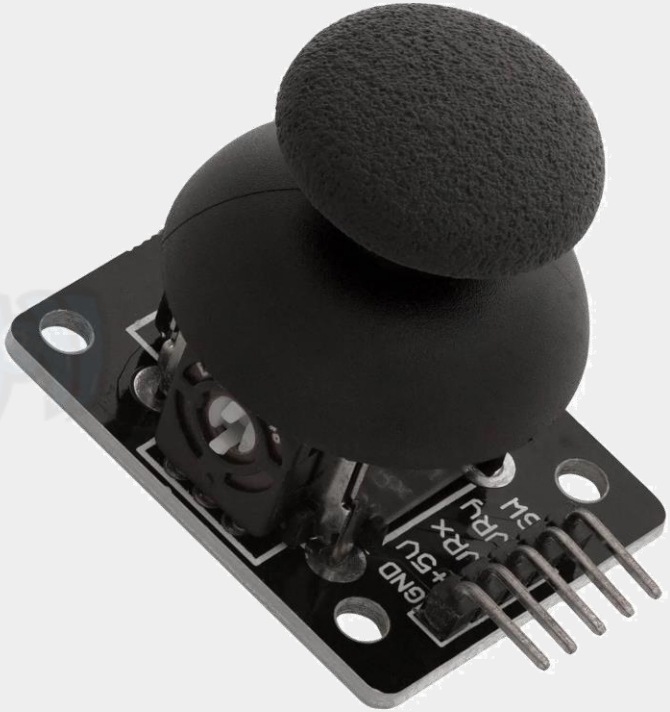
The Basics



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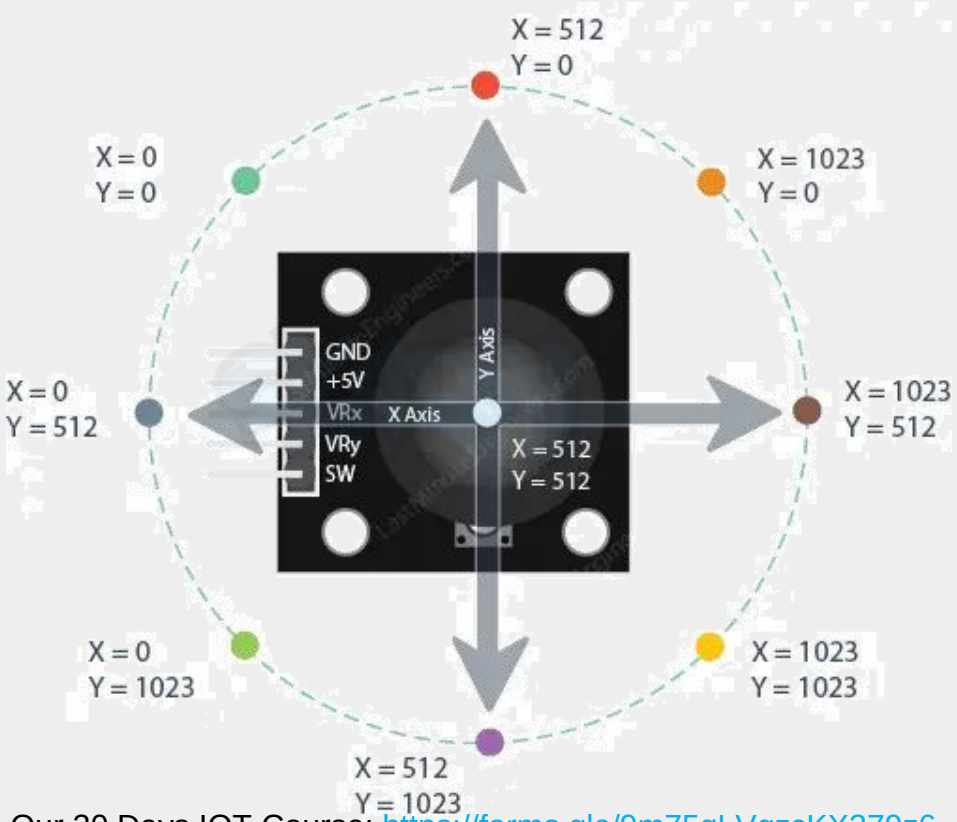
Pinout





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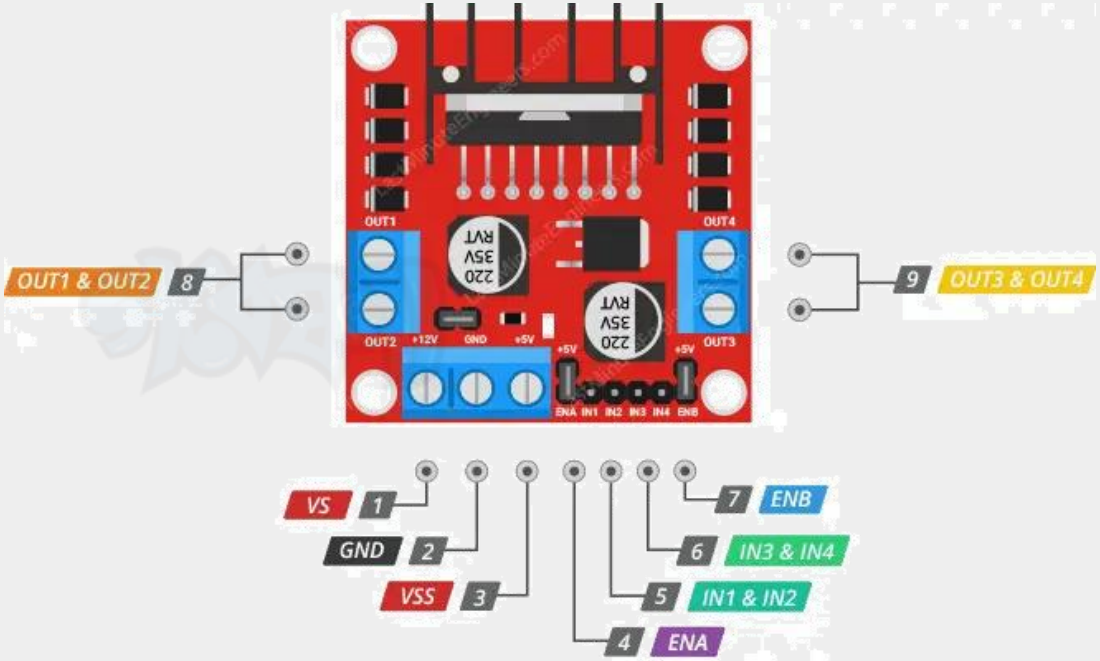
Reading Values



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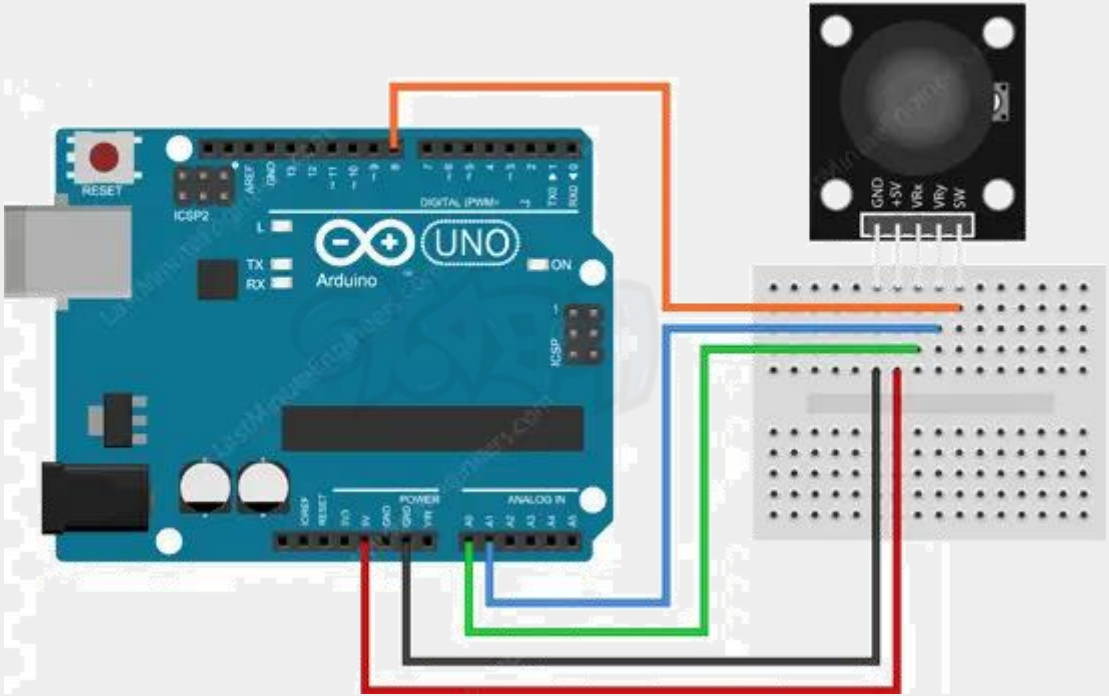
Motor Driver

- **Model: L298N**



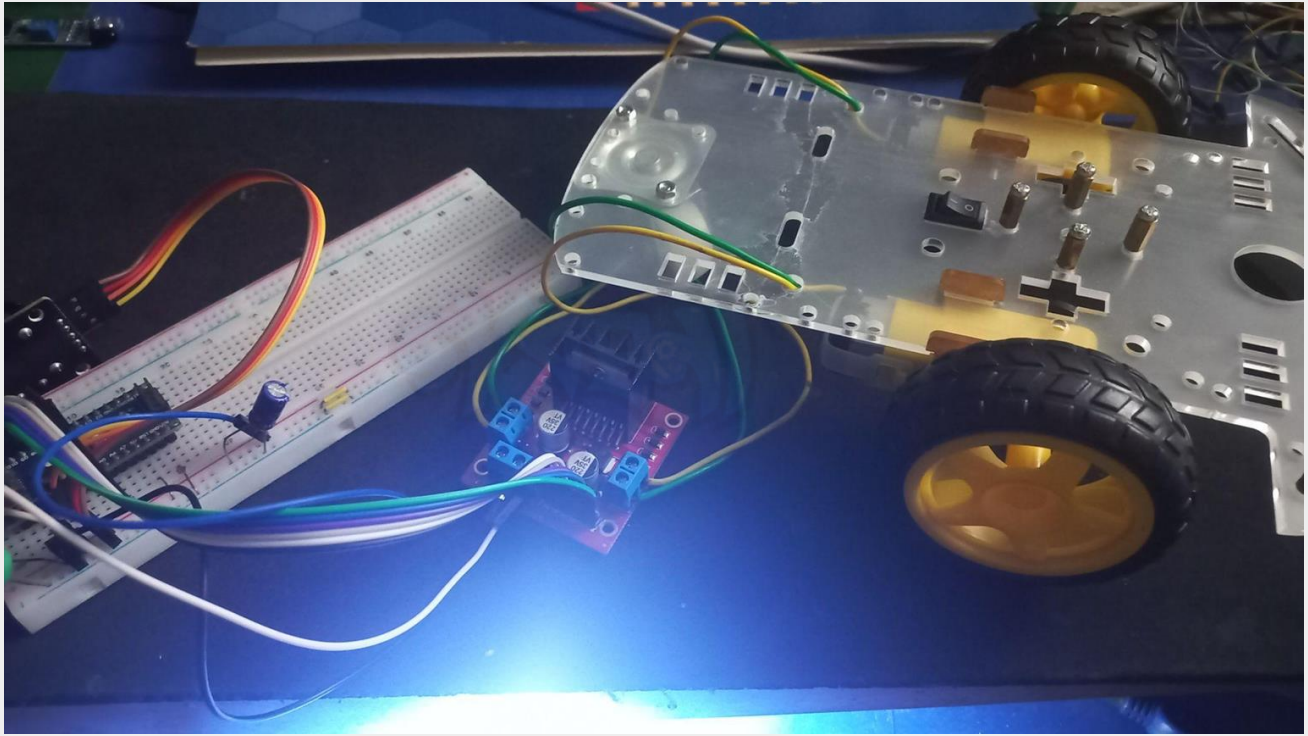
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Sample Circuit



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Sample Circuit



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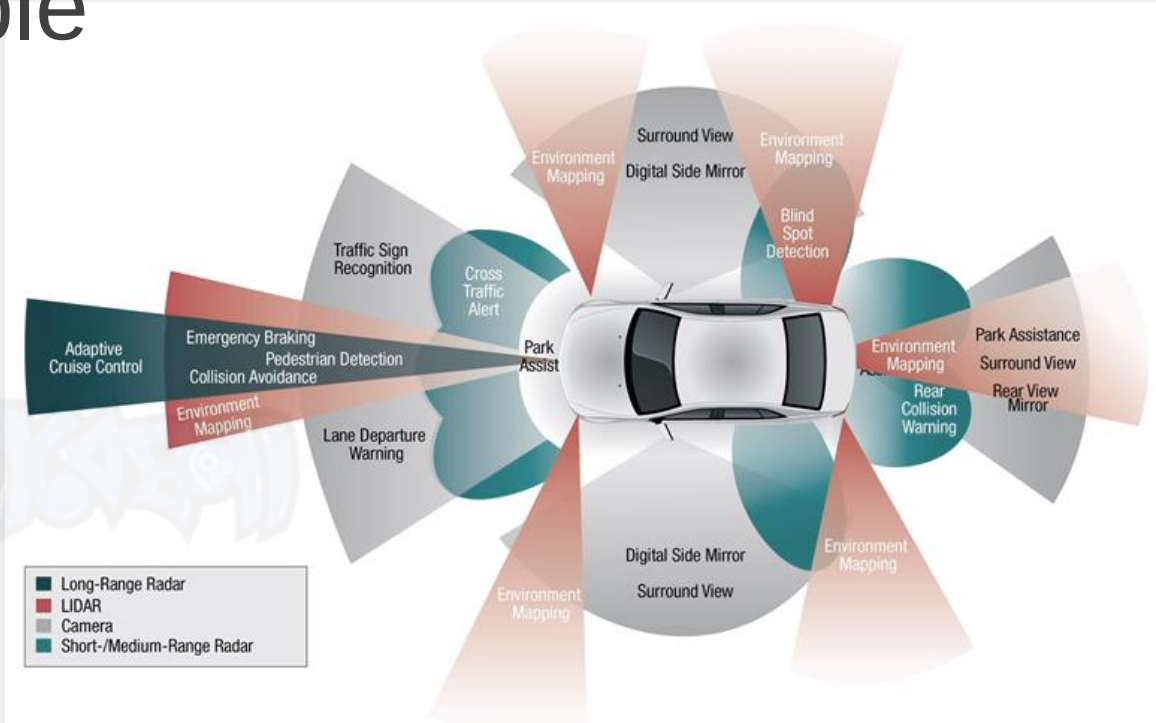
Autonomous Obstacle Avoidance Robot

What is it?

- Robot navigates autonomously
- Sensors detect obstacles
- Real-time decision-making
- Wide range of applications
- Enhanced safety and efficiency.

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Working Principle



How it works?

- Sensors detect surroundings.
- Computer processes data.
- Vehicle controls movement.
- Machine learning improves performance.
- Autonomous cars enhance safety and efficiency.

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How to build a simple one?

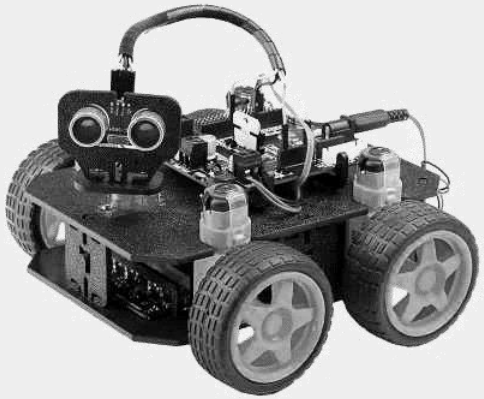


Necessary Steps:

1. **Gather necessary components:** Arduino microcontroller, sensors (e.g., ultrasonic, infrared), motors, wheels, power supply, and breadboard.
2. **Connect components:** Assemble the circuit using the breadboard, connecting the sensors, motors, and power supply to the Arduino.
3. **Write code:** Use Arduino programming language to create a program that reads sensor data, determines the distance to obstacles, and controls the motors to avoid them.
4. **Test & calibrate:** Upload the code to the Arduino, test the robot's obstacle avoidance behavior, and adjust the code as needed.
5. **Refine and improve:** Continuously experiment with different sensor configurations, algorithms, and motor control techniques to enhance the robot's performance.

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How to build a simple one?

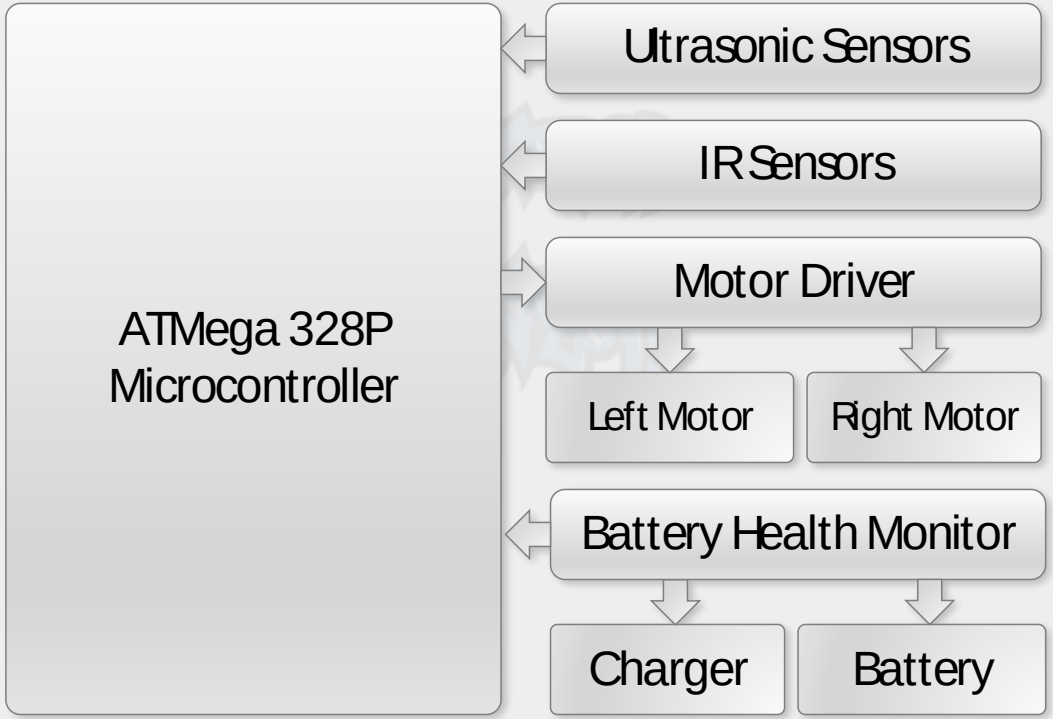


The Algorithm:

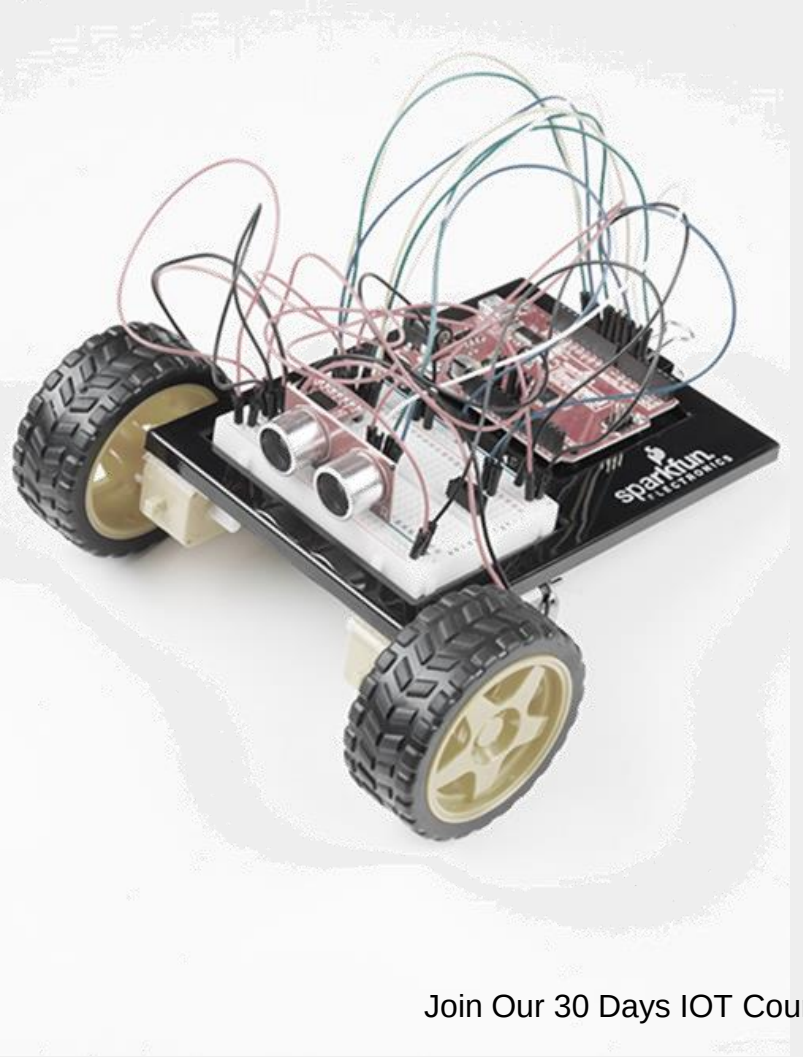
- **Initialize** the sensors.
- Read **ultrasonic sensors** to get obstacle distance.
- **Compare** the distance with threshold distance.
- If it is **less or equal**, that means obstacle is present.
- Then check for **IR sensor** to detect free space.
- If movable distance is available, then **turn left/right**.
- Make a **U-turn**.
- Check again, **run** normally.
- **Repeat.**

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System Architecture



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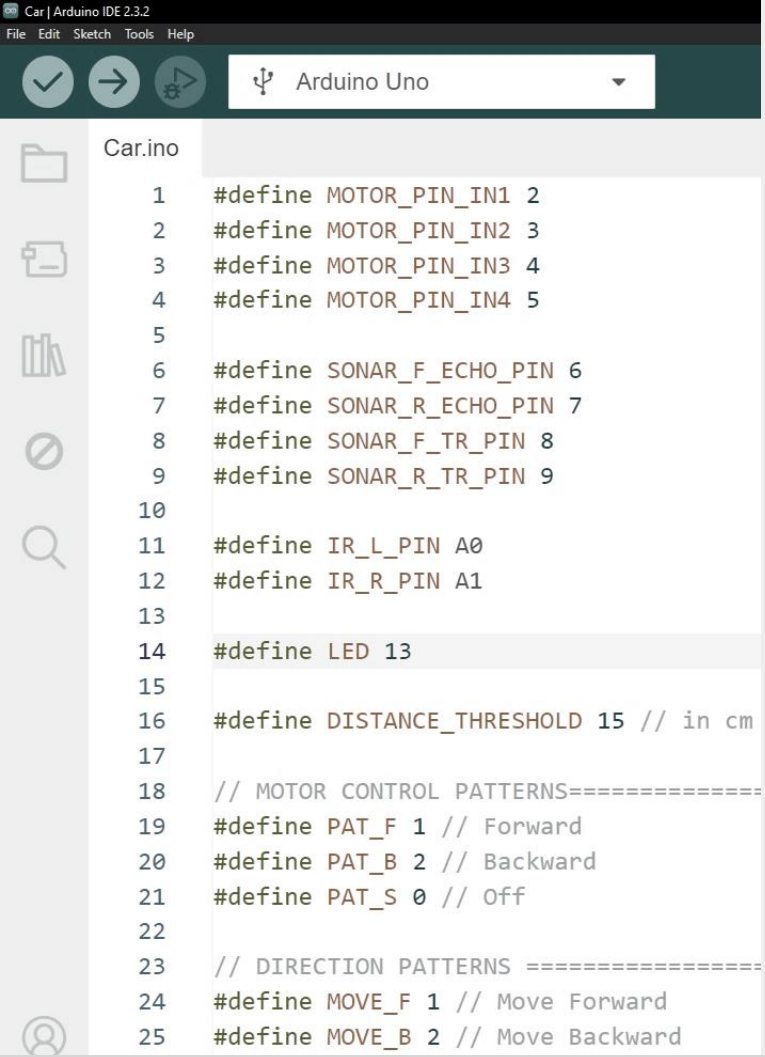


Components

Needed items:

- Arduino uno/nano: ATMega328p microcontroller
- Motor Controller: L298N
- Ultrasonic Sensor: HC-SR04 (x2)
- IR Sensor (x2)
- 12v DC motor (x2)
- 7v/9v/12v battery pack
- Battery Charger
- Car chassis (Plexiglass based)
- Jumper Wires
- LED, Breadboard

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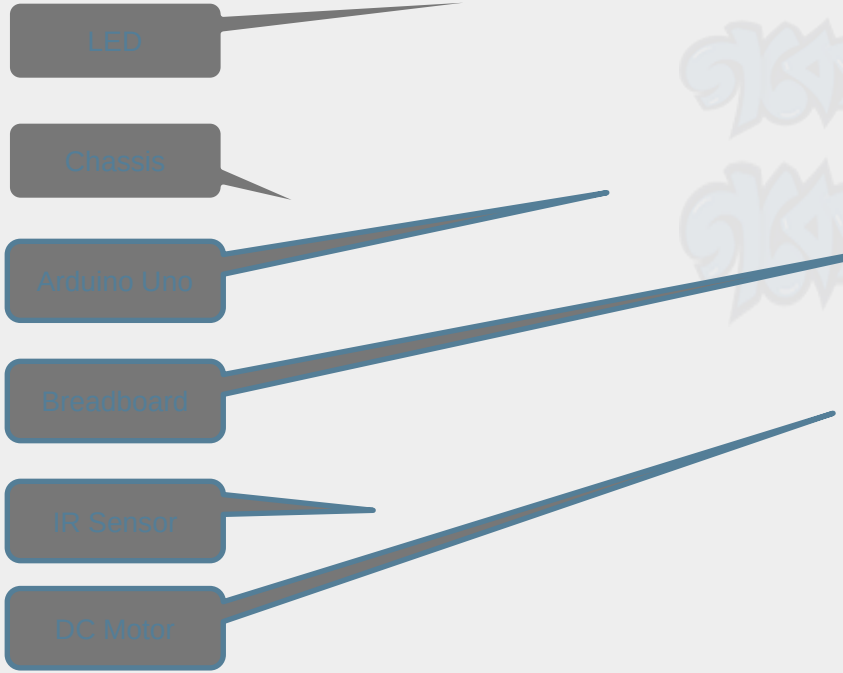


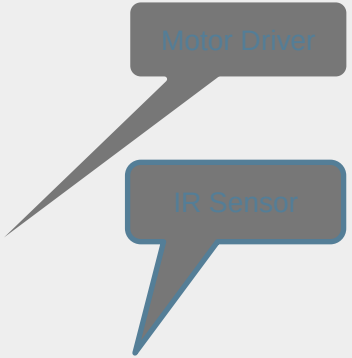
Sample Circuit

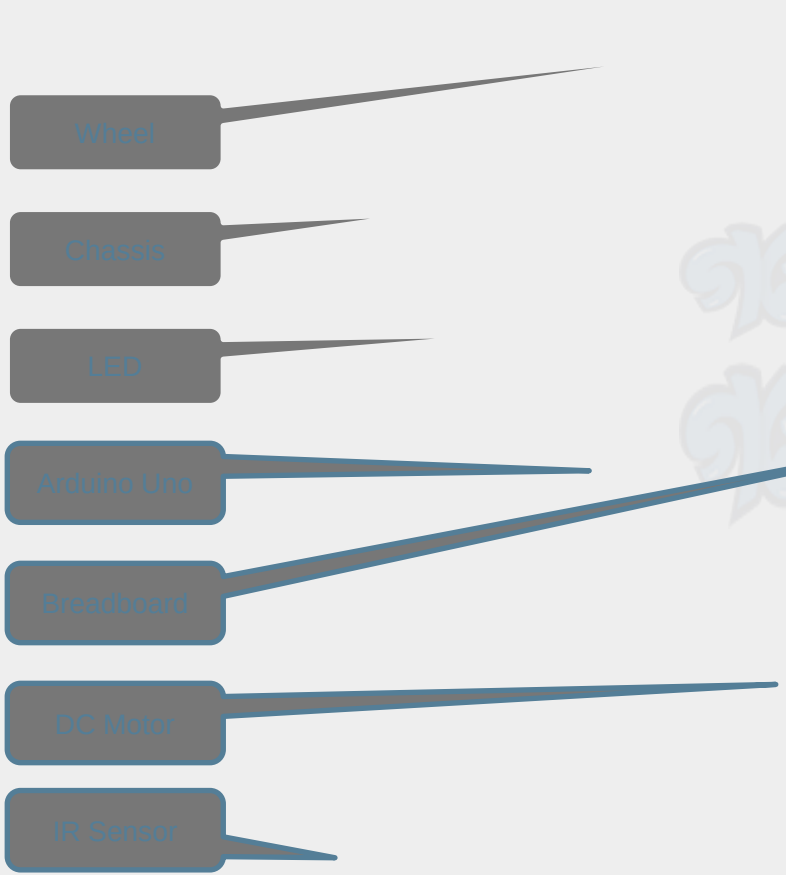
Pinouts:

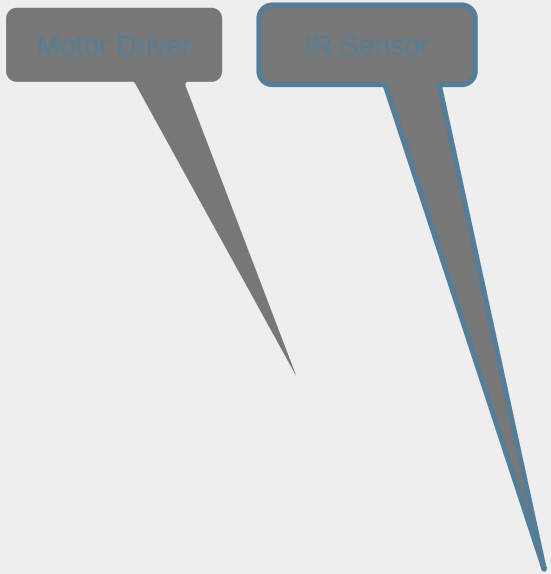
Arduino End	Component End
D2	Motor Driver IN1
D3	Motor Driver IN2
D4	Motor Driver IN3
D5	Motor Driver IN4
D6	Sonar Front Echo
D7	Sonar Rear Echo
D8	Sonar Front Trigger
D9	Sonar Rear Trigger
D13	LED
A0	IR Sensor Left
A1	IR Sensor Right

The Prototype









The Prototype

Any question?



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Thank you

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Timekeeping with Arduino

Why?

- When you need precise time in your project.
- When time is crucial.
- Datalogging purpose.
- Perform time based event/jobs with Arduino.

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Real time clock



How it works?

- Keeps time with high accuracy.
- Battery powered.
- Volatile memory.
- Interface with microcontrollers.
- Low powered device.

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Real time clock

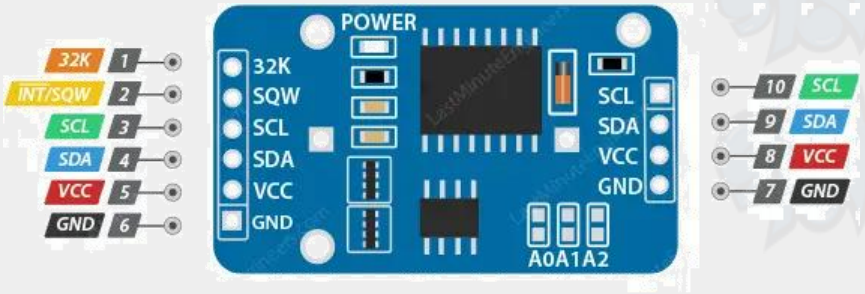


Specifications:

- Model: DS3231
- Battery backup: 3v max
- Power rating: 2.3-5.5v 300µA (typ.)
- I2c interface
- Can store date, time, leap year, week name
- Built in temperature sensor

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Sample Circuit



Pinout:

Arduino End	RTC
VCC	5V
GND	GND
SDA	SDA
SCL	SCL

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The Circuit

The Circuit

Any question?



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Thank you

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Introduction to ESP32



Overview:

- A high-performance microcontroller board.
- Built in Hall effect, Touch, Temperature sensor.
- Offers Wi-Fi, Bluetooth, i2c, i2s, spi, can-bus and many more protocols.
- All pins are PWM supported.
- 12Bit ADC & DAC support.
- Dual Core CPU.

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Introduction to ESP32



Specifications:

- 2.2v-3.6v operation. 600mA on board LDO.
- Up to 240MHz Clock Freq. based dual core CPU.
- 520kB internal SRAM, 4MB external flash.
- 15 ADC channels (12 bit), 2 UART with IrDA support.
- 25 PWM IO, 2 8 bit DAC, 3 SPI, 1 i2C, 9 Touch pads.
- Price: $6 - $9. (similar to ESP8266).

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Available Models



ESP32 DevKit V1

Because of its immense popularity, this is the best board for beginners. It comes with everything you need to get started with ESP32. And this is what we will use for our experiments.

ESP32 OLED

Compared to the previous board, this one includes an OLED display, making it a neat addition for Internet of Things applications. Additionally, it has a u.FL connector for an external antenna and a lipo battery connector for creating battery-operated projects.



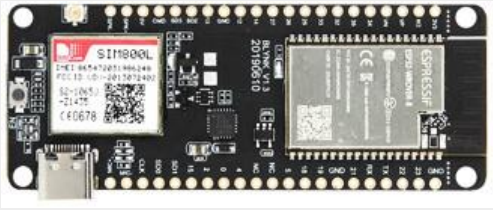


ESP32 Cam

Built-in OV2640 2MP camera and microSD card slot make this one of the most interesting boards available; it's ideal for IoT projects that require a camera with advanced functions like image tracking and recognition.

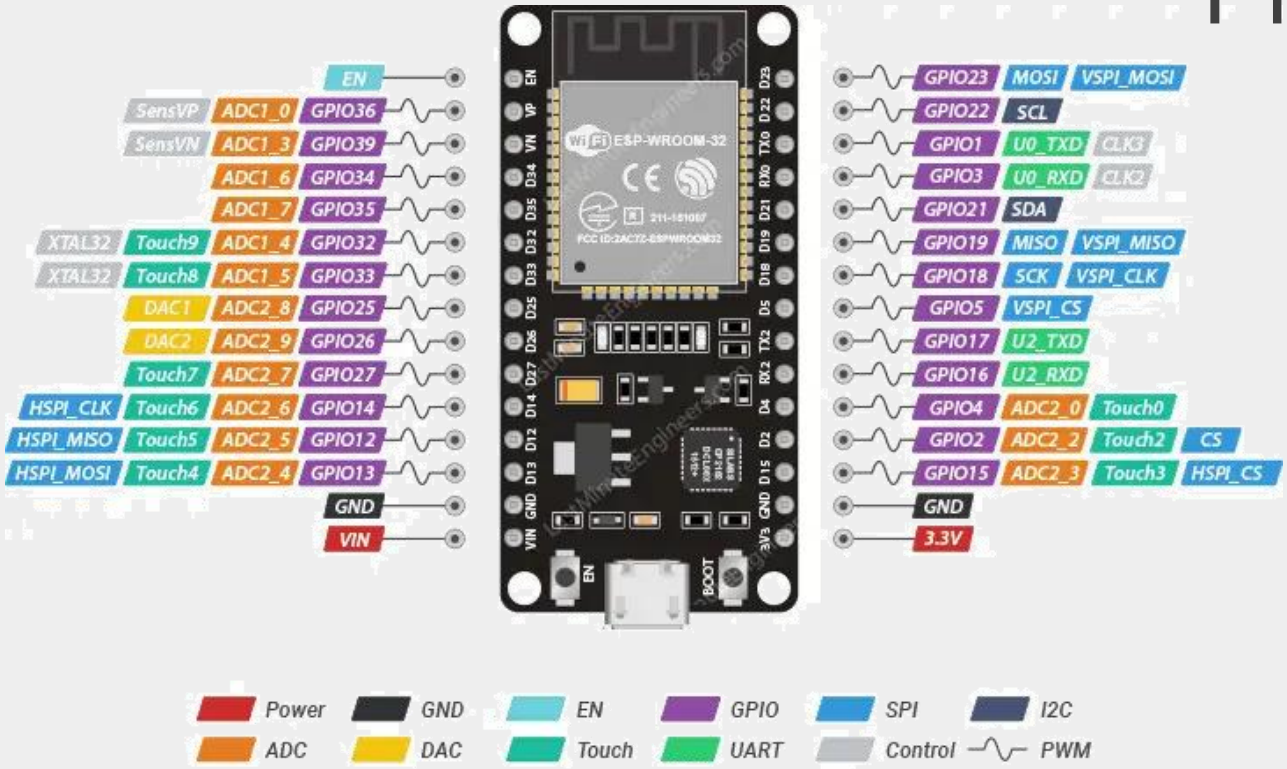
ESP32 SIM800L

The ESP32 SIM800L TTGO T-Call board allows you to connect to the Internet via a SIM card data plan and push information to the cloud, without the need for Wi-Fi.



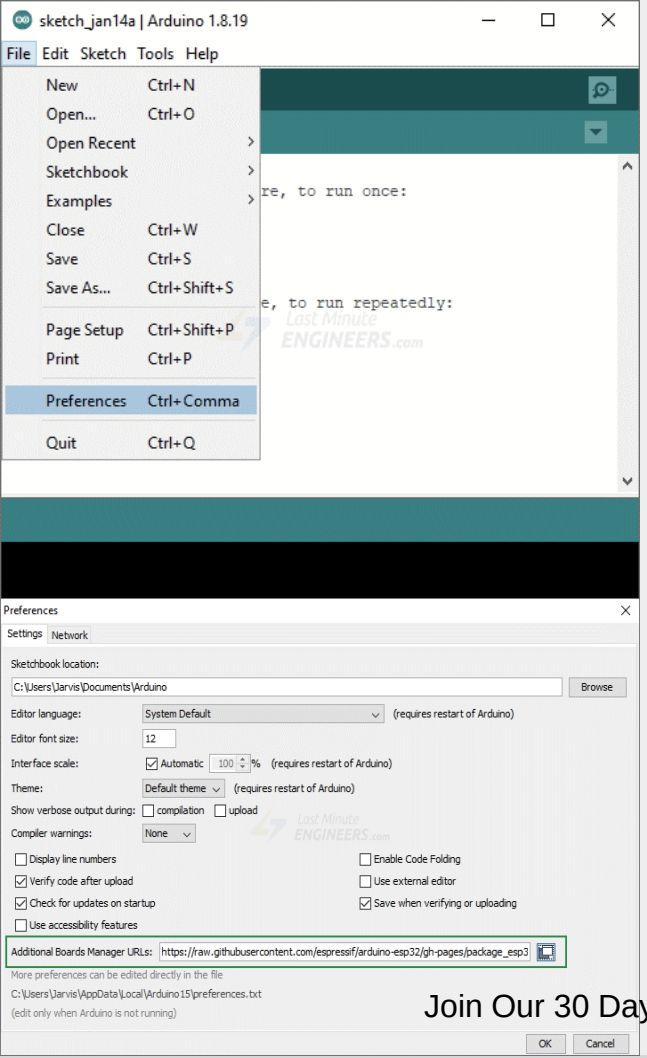
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PinOut



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Installation



Necessary Steps:

1. **Install Arduino IDE:** Download and install Arduino IDE 2.x from Arduino.cc.
2. **Configure the IDE:** Go to File/Preferences. Fill in the “Additional Board Manager URLs” field with the following. This will include support for both ESP32 and ESP8266 boards.
   https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json,
   http://arduino.esp8266.com/stable/package_esp8266com_index.json
3. **Install ESP32 Board:** Now navigate to Tools > Board > Boards Manager…Filter your search by entering ‘esp32‘. Look for ESP32 by Espressif Systems. Click on that entry, and then choose Install.
4. **Select board:** Now select your board in the Tools > Board menu (in our case, it’s the DOIT ESP32 DEVKIT V1. If you are unsure which board you have, select ESP32 Dev Module.

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Interfacing with Internal Temperature Sensor

Purpose:

System health check.

Accessing Method:

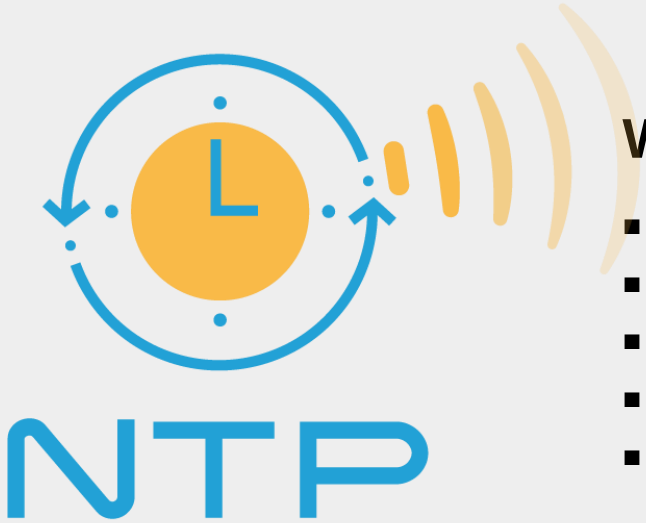
`temperatureRead();`

Return Type:

Float

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Network Time Protocol (NTP)

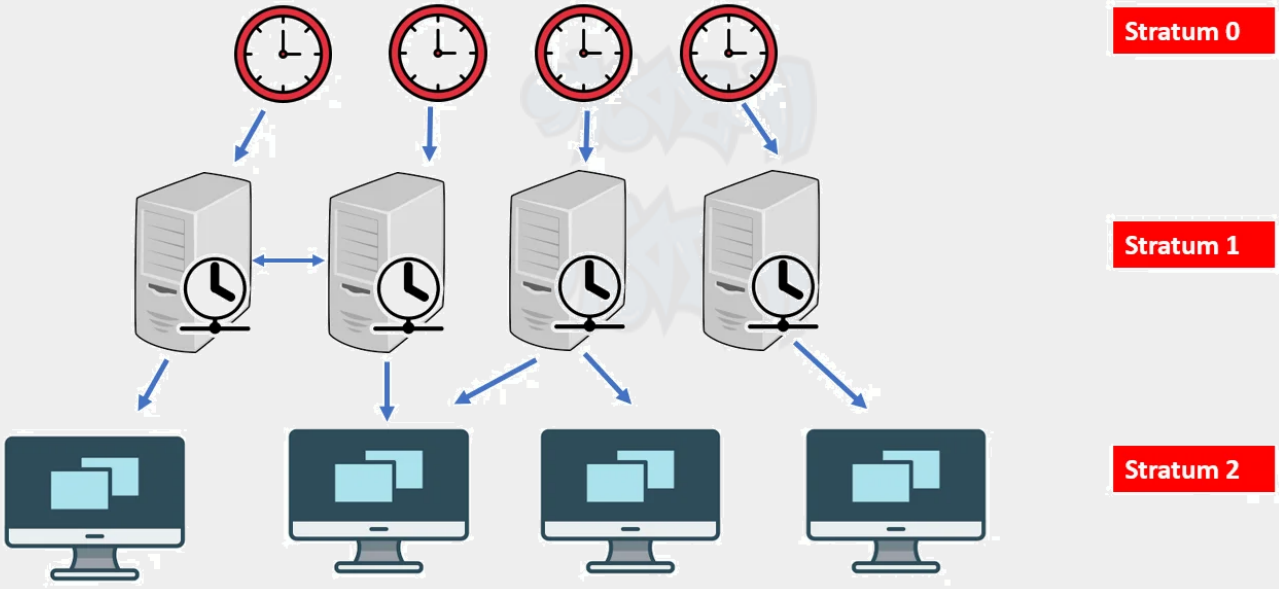


Why NTP?

- NTP is a protocol for time synchronization.
- It relies on a hierarchy of time servers.
- Primary servers are connected to atomic clocks.
- Clients query servers to obtain precise time.
- NTP is essential for various time-sensitive applications.

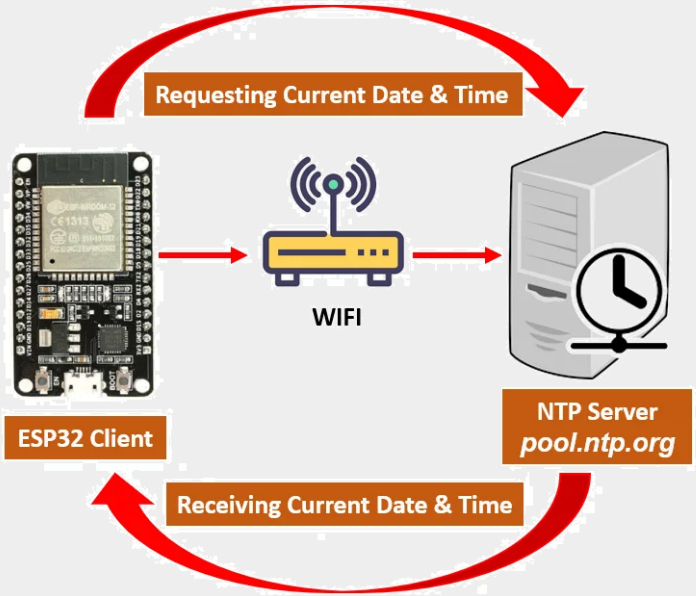
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Network Time Protocol (NTP)



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Network Time Protocol (NTP)



How it works?

- Hierarchy: NTP creates a hierarchy of time servers.
- Packets: Clients send packets to servers.
- Calculation: Servers calculate time differences.
- Adjustment: Servers send time adjustments to clients.
- Synchronization: NTP periodically exchanges packets for accurate time.

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